



FATIMA JINNAH WOMEN UNIVERSITY

Submitted To:

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Section:

5A

CLOUD COMPUTING

Lab No.12

Task1

```
@23-22411-027-max →/workspaces/lab-12 (main) $ mkdir -p ~/Lab12  
cd ~/Lab12
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  bash - Lab12 + v [ ] [ ] ... | [ ] [ ] X
```

```
@23-22411-027-max →~/Lab12 $ touch main.tf variables.tf outputs.tf locals.tf terraform.tfvars en  
try-script.sh  
ls -la  
-rw-r--r-- 1 codespace codespace 0 Jan 3 04:12 locals.tf  
-rw-r--r-- 1 codespace codespace 0 Jan 3 04:12 main.tf  
-rw-r--r-- 1 codespace codespace 0 Jan 3 04:12 outputs.tf  
-rw-r--r-- 1 codespace codespace 0 Jan 3 04:12 terraform.tfvars  
-rw-r--r-- 1 codespace codespace 0 Jan 3 04:12 variables.tf  
@23-22411-027-max →~/Lab12 $ nano variables.tf  
@23-22411-027-max →~/Lab12 $
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  nano - Lab12 +
```

```
GNU nano 7.2 variables.tf *  
variable "vpc_cidr_block" {}  
variable "subnet_cidr_block" {}  
variable "availability_zone" {}  
variable "env_prefix" {}  
variable "instance_type" {}  
variable "public_key" {}  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS nano - Lab12 + -

```
GNU nano 7.2 outputs.tf *
output "aws_instance_public_ip" {
  value = aws_instance.myapp-server.public_ip
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS nano - Lab12 + -

```
GNU nano 7.2 locals.tf *
locals {
  my_ip = "${chomp(data.http.my_ip.response_body)}/32"
}

data "http" "my_ip" {
  url = "https://icanhazip.com"
```

Help Write Out Where Is Cut Execute Loca

GNU nano 7.2 terraform.tfvars *

```
vpc_cidr_block      = "10.0.0.0/16"
subnet_cidr_block   = "10.0.10.0/24"
availability_zone    = "me-central-1a"
env_prefix           = "dev"
instance_type        = "t3.micro"
public_key           = "~/ssh/id_ed25519.pub"
```

Help Write Out Where Is Cut Execute Location
 Exit Read File Replace Paste Justify Go To Lin

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS nano - Lab12 + v [ ] [ ] ... | [ ]
GNU nano 7.2 main.tf *
from_port = 0
to_port = 0
protocol = "-1"
cidr_blocks = ["0.0.0.0/0"]
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS nano - Lab12 + v [ ] [ ] ... | [ ]
GNU nano 7.2 entry-script.sh *
#!/bin/bash
set -e
yum update -y
yum install -y nginx
systemctl start nginx
systemctl enable nginx

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash - Lab12 + v [ ] [ ] ... | [ ]
@23-22411-027-max → ~/Lab12 $ terraform init

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@23-22411-027-max → ~/Lab12 $
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  bash - Lab12  + -  [ ] [X] ...

@23-22411-027-max →~/Lab12 $ terraform apply -auto-approve
aws_instance.myapp-server: Still creating... [00m10s elapsed]
aws_instance.myapp-server: Creation complete after 13s [id=i-0c88ed25e94d07442]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

aws_instance_public_ip = "51.112.47.28"
@23-22411-027-max →~/Lab12 $ █

aws_instance_public_ip = 51.112.47.28
@23-22411-027-max →~/Lab12 $ terraform output
aws_instance_public_ip = "51.112.47.28"
@23-22411-027-max →~/Lab12 $ █

⚠ Not secure 51.112.47.28
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

```
@23-22411-027-max →~/Lab12 $ terraform destroy
aws_default_security_group.default_sg: Destroying... [id=sg-023d963e80a292cd5]
aws_default_security_group.default_sg: Destruction complete after 0s
aws_key_pair.ssh-key: Destruction complete after 1s
aws_subnet.myapp_subnet_1: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0352afea926b4df4f]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
@23-22411-027-max →~/Lab12 $ █
```

TASK 2

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS sudo - Lab12 + v [ ] [ ] ... | [ ] [X]
GNU nano 7.2 main.tf *
resource "aws_instance" "myapp-server" {
  ami           = "ami-05524d6658fcf35b6"
  instance_type = var.instance_type
  subnet_id     = aws_subnet.myapp_subnet_1.id
  security_groups = [aws_default_security_group.default_sg.id]
  availability_zone = var.availability_zone
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^A Replace ^U Paste ^J Justify ^_ Go To Line

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash - Lab12 + v [ ] [ ] ... | [ ] [X]
@23-22411-027-max → ~/Lab12 $ terraform apply -auto-approve
aws_instance.myapp-server (remote-exec): Verifying : nginx-mimety 7/7

aws_instance.myapp-server (remote-exec): Installed:
aws_instance.myapp-server (remote-exec): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
aws_instance.myapp-server (remote-exec): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): libunwind-1.4.0-5.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): nginx-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch
aws_instance.myapp-server (remote-exec): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

aws_instance.myapp-server (remote-exec): Complete!
aws_instance.myapp-server (remote-exec): Created symlink /etc/systemd/system/multi-user.target.
nts/nginx.service → /usr/lib/systemd/system/nginx.service.
aws_instance.myapp-server: Creation complete after 36s [id=i-064be5ecb06d4d7bd]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

aws_instance_public_ip = "158.252.85.83"
@23-22411-027-max → ~/Lab12 $
```

```
@23-22411-027-max → ~/Lab12 $ terraform apply -auto-approve
```

```
aws_instance.myapp-server (remote-exec): Verifying      : nginx-filesy 6/7  
aws_instance.myapp-server (remote-exec): Verifying      : nginx-mimety 7/7
```

```
aws_instance.myapp-server (remote-exec): Installed:
```

```
aws_instance.myapp-server (remote-exec): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  
aws_instance.myapp-server (remote-exec): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64  
aws_instance.myapp-server (remote-exec): libunwind-1.4.0-5.amzn2023.0.3.x86_64  
aws_instance.myapp-server (remote-exec): nginx-1:1.28.0-1.amzn2023.0.2.x86_64  
aws_instance.myapp-server (remote-exec): nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64  
aws_instance.myapp-server (remote-exec): nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch  
aws_instance.myapp-server (remote-exec): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch
```

```
aws_instance.myapp-server (remote-exec): Complete!
```

```
aws_instance.myapp-server (remote-exec): Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service →  
/usr/lib/systemd/system/nginx.service.
```

```
aws_instance.myapp-server: Creation complete after 34s [id=i-089e681d88a099163]
```

```
Apply complete! Resources: 4 added, 1 changed, 3 destroyed.
```

```
Outputs:
```

```
aws_instance_public_ip = "51.112.47.72"
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

```
@23-22411-027-max →~/Lab12 $ terraform destroy
```

```
aws_route_table_association.myapp_rta: Destroying... [id=rtbassoc-0d818666dbbd44475]
aws_instance.myapp-server: Destroying... [id=i-089e681d88a099163]
aws_route_table_association.myapp_rta: Destruction complete after 0s
aws_route_table.myapp_rt: Destroying... [id=rtb-0e3fab7943c7e8289]
aws_route_table.myapp_rt: Destruction complete after 1s
aws_internet_gateway.myapp_igw: Destroying... [id=igw-036b623f9a5c2f996]
aws_instance.myapp-server: Still destroying... [id=i-089e681d88a099163, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-036b623f9a5c2f996, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Destruction complete after 17s
aws_instance.myapp-server: Still destroying... [id=i-089e681d88a099163, 00m20s elapsed]
aws_instance.myapp-server: Destruction complete after 30s
aws_key_pair.ssh-key: Destroying... [id=serverkey]
aws_subnet.myapp_subnet_1: Destroying... [id=subnet-06f3f83ef9df336b9]
aws_security_group.myapp_sg: Destroying... [id=sg-0ab921dba10116ba3]
aws_key_pair.ssh-key: Destruction complete after 0s
aws_subnet.myapp_subnet_1: Destruction complete after 1s
aws_security_group.myapp_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-042de7e60aedb87f5]
aws_vpc.myapp_vpc: Destruction complete after 1s
```

```
Destroy complete! Resources: 8 destroyed.
```

```
@23-22411-027-max →~/Lab12 $
```

TASK3

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS sudo - Lab12 + - [] [] ... [] [] X

GNU nano 7.2 main.tf *

File provisioner – upload script
provisioner "file" {
 source = "./entry-script.sh"
 destination = "/home/ec2-user/entry-script-on-ec2.sh"
}

Remote-exec provisioner – run script on EC2
provisioner "remote-exec" {
 inline = [
 "sudo chmod +x /home/ec2-user/entry-script-on-ec2.sh",
 "sudo /home/ec2-user/entry-script-on-ec2.sh"
]
}

Local-exec provisioner – log info locally
provisioner "local-exec" {
 command = <<-EOF
 echo Instance \${self.id} with public IP \${self.public_ip} has been created
 EOF
}

tags = {
 Name = "\${var.env_prefix}-ec2-instance"
}
}

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

bash - Lab12 + v [icon] [icon] ... | [icon] [icon] X

@23-22411-027-max →~/Lab12 \$ terraform init

terraform apply -auto-approve

aws_instance.myapp-server (remote-exec): Verifying : nginx-filesy 6/7

aws_instance.myapp-server (remote-exec): Verifying : nginx-mimety 7/7

aws_instance.myapp-server (remote-exec): Installed:

aws_instance.myapp-server (remote-exec): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch

aws_instance.myapp-server (remote-exec): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64

aws_instance.myapp-server (remote-exec): libunwind-1.4.0-5.amzn2023.0.3.x86_64

aws_instance.myapp-server (remote-exec): nginx-1:1.28.0-1.amzn2023.0.2.x86_64

aws_instance.myapp-server (remote-exec): nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64

aws_instance.myapp-server (remote-exec): nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch

aws_instance.myapp-server (remote-exec): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

aws_instance.myapp-server (remote-exec): Complete!

aws_instance.myapp-server (remote-exec): Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.

aws_instance.myapp-server: Provisioning with 'local-exec'...

aws_instance.myapp-server (local-exec): Executing: ["/bin/sh" "-c" "echo Instance i-0f04843e6faa8cc7d with public IP 3.28.135.6 has been created\n"]

aws_instance.myapp-server (local-exec): Instance i-0f04843e6faa8cc7d with public IP 3.28.135.6 has been created

aws_instance.myapp-server: Creation complete after 1m1s [id=i-0f04843e6faa8cc7d]

Apply complete! Resources: 1 added, 2 changed, 0 destroyed.

Outputs:

aws_instance_public_ip = "3.28.135.6"

● @23-22411-027-max →~/Lab12 \$ terraform output

aws_instance_public_ip = "3.28.135.6"

○ @23-22411-027-max →~/Lab12 \$ █

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

```
@23-22411-027-max →~/Lab12 $ terraform destroy
```

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
aws_route_table_association.myapp_rta: Destroying... [id=rtbassoc-0f2094bcea12cbcf6]
aws_instance.myapp-server: Destroying... [id=i-0f04843e6faa8cc7d]
aws_route_table_association.myapp_rta: Destruction complete after 1s
aws_route_table.myapp_rt: Destroying... [id=rtb-09b1a4a1d88fdaa4b]
aws_route_table.myapp_rt: Destruction complete after 0s
aws_internet_gateway.myapp_igw: Destroying... [id=igw-0a8dbb2ec2adaa766]
aws_instance.myapp-server: Still destroying... [id=i-0f04843e6faa8cc7d, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0a8dbb2ec2adaa766, 00m10s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-0f04843e6faa8cc7d, 00m20s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0a8dbb2ec2adaa766, 00m20s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-0f04843e6faa8cc7d, 00m30s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0a8dbb2ec2adaa766, 00m30s elapsed]
aws_internet_gateway.myapp_igw: Destruction complete after 38s
aws_instance.myapp-server: Still destroying... [id=i-0f04843e6faa8cc7d, 00m40s elapsed]
aws_instance.myapp-server: Destruction complete after 41s
aws_subnet.myapp_subnet_1: Destroying... [id=subnet-0a1fb953af1d0679f]
aws_key_pair.ssh-key: Destroying... [id=serverkey]
aws_security_group.myapp_sg: Destroying... [id=sg-0248bdd0bf31447a5]
aws_key_pair.ssh-key: Destruction complete after 0s
aws_subnet.myapp_subnet_1: Destruction complete after 0s
aws_security_group.myapp_sg: Destruction complete after 0s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0cda72e87363cd088]
aws_vpc.myapp_vpc: Destruction complete after 1s
```

Destroy complete! Resources: 8 destroyed.

```

@23-22411-027-max →~/Lab12 $ ls -l main.tf
-rw-r--r-- 1 @23-22411-027-max 4096 Nov 14 14:54 main.tf
cat main.tf
key_name    = "serverkey"
public_key  = file(var.public_key)
}

# EC2 Instance with remote-exec
resource "aws_instance" "myapp-server" {
  ami                = "ami-05524d6658fcf35b6"
  instance_type      = var.instance_type
  subnet_id          = aws_subnet.myapp_subnet_1.id
  security_groups    = [aws_security_group.myapp_sg.id]
  associate_public_ip_address = true
  key_name            = aws_key_pair.ssh-key.key_name
  availability_zone   = var.availability_zone

  user_data = file("./entry-script.sh")

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}

```

TASK4

```

Destroy Complete!
resources: 0 destroyed.
@23-22411-027-max →~/Lab12 $ sudo nano main.t
@23-22411-027-max →~/Lab12 $ mkdir -p modules
touch modules/subnet/main.tf
touch modules/subnet/variables.tf
touch modules/subnet/outputs.tf
@23-22411-027-max →~/Lab12 $

```

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  nano - Lab12 + v
GNU nano 7.2 modules/subnet/variables.tf *
variable "vpc_id" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
variable "default_route_table_id" {}

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS nano - Lab12

GNU nano 7.2

modules/subnet/main.tf *

```
# Route Table
resource "aws_default_route_table" "main_rt" {
  default_route_table_id = var.default_route_table_id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.myapp_igw.id
  }

  tags = {
    Name = "${var.env_prefix}-rt"
  }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

nano - Lab12 + v [] [] ... |

GNU nano 7.2

modules/subnet/outputs.tf *

```
output "subnet" {
  value = aws_subnet.myapp_subnet_1
}
```

```
GNU nano 7.2 main.tf *
type      = "ssh"
user      = "ec2-user"
private_key = file(var.private_key)
host      = self.public_ip
}

provisioner "file" {
  source      = "./entry-script.sh"
  destination = "/home/ec2-user/entry-script-on-ec2.sh"
}

provisioner "remote-exec" {
  inline = [
    "sudo chmod +x /home/ec2-user/entry-script-on-ec2.sh",
    "sudo /home/ec2-user/entry-script-on-ec2.sh"
  ]
}

provisioner "local-exec" {
  command = <<-EOF
    echo Instance ${self.id} with public IP ${self.public_ip} has been created
  EOF
}

tags = {
  Name = "${var.env_prefix}-ec2-instance"
}
```

● @23-22411-027-max →~/Lab12 \$ sudo nano main.tf

● @23-22411-027-max →~/Lab12 \$ terraform init

Initializing the backend...

Initializing modules...

- myapp-subnet in modules/subnet

Initializing provider plugins...

- Reusing previous version of hashicorp/aws from the dependency lock file

- Reusing previous version of hashicorp/http from the dependency lock file

- Using previously-installed hashicorp/aws v6.27.0

- Using previously-installed hashicorp/http v3.5.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

○ @23-22411-027-max →~/Lab12 \$ █


```
@23-22411-027-max →~/Lab12 $ terraform apply -auto-approve
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ aws_instance_public_ip = (known after apply)
aws_instance.myapp-server: Creating...
aws_instance.myapp-server: Still creating... [00m10s elapsed]
aws_instance.myapp-server: Creation complete after 13s [id=i-0194108d581e4f516]
```

Warning: Value for undeclared variable

The root module does not declare a variable named "vpc_cidr_block" but a value was found : file "terraform.tfvars". If you meant to use this value, add a "variable" block to the configuration.

To silence these warnings, use TF_VAR_... environment variables to provide certain "global" settings to all configurations in your organization. To reduce the verbosity of these warnings, use the -compact-warnings option.

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

```
aws_instance_public_ip = "3.29.129.18"
> @23-22411-027-max →~/Lab12 $ terraform output
bash: terraform_output: command not found
> @23-22411-027-max →~/Lab12 $ terraform output
aws_instance_public_ip = "3.29.129.18"
> @23-22411-027-max →~/Lab12 $ █
```

⚠ Not secure 3.29.129.18

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

TASK5

GNU nano 7.2

main.tf *

```
.
: Use the webserver module (Task 5)
module "myapp-webserver" {
  source = "../modules/webserver"
  env_prefix      = var.env_prefix
  instance_type   = var.instance_type
  availability_zone = var.availability_zone
  public_key      = var.public_key
  my_ip           = local.my_ip
  vpc_id          = aws_vpc.myapp_vpc.id
}
```

GNU nano 7.2

modules/webserver/variables.tf *

```
variable "env_prefix" {}
variable "instance_type" {}
variable "availability_zone" {}
variable "public_key" {}
variable "my_ip" {}
variable "vpc_id" {}
variable "subnet_id" {}
variable "script_path" {}
variable "instance_suffix" {}
|
```

^G Help
^X Exit

^O Write Out
^R Read File

^W Where Is
^_ Replace

^K Cut
^U Paste

^T Execute
^J Justify

^C Location
^_ Go To Line

[Preview] README.md

main.tf U X

Lab12 > modules > webserver > main.tf

```
1  ✓ resource "aws_security_group" "web_sg" {
2    vpc_id      = var.vpc_id
3    name        = "${var.env_prefix}-web-sg-${var.instance_suffix}"
4    description = "Security group for web server allowing HTTP, HTTPS and
5
6  ✓ ingress {
7    from_port = 22
8    to_port   = 22
9    protocol  = "tcp"
10   cidr_blocks = [var.my_ip]
```

[Preview] README.md

main.tf U

outputs.tf U X

Lab12 > modules > webserver > outputs.tf

```
1  output "aws_instance" {
2    value = aws_instance.myapp-server
3  }
4  |
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash - Lab12 + - [ ] [ ] ... [ ] [ ] X
@23-22411-027-max → ~/Lab12 $ terraform init
terraform apply -auto-approve
aws_instance.myapp-server (remote-exec): Verifying : nginx-filesy 6/7
aws_instance.myapp-server (remote-exec): Verifying : nginx-mimety 7/7

aws_instance.myapp-server (remote-exec): Installed:
aws_instance.myapp-server (remote-exec): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
aws_instance.myapp-server (remote-exec): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): libunwind-1.4.0-5.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): nginx-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch
aws_instance.myapp-server (remote-exec): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

aws_instance.myapp-server (remote-exec): Complete!
aws_instance.myapp-server (remote-exec): Created symlink /etc/systemd/system/multi-user.target.wa
nts/nginx.service → /usr/lib/systemd/system/nginx.service.
aws_instance.myapp-server: Provisioning with 'local-exec'...
aws_instance.myapp-server (local-exec): Executing: ["/bin/sh" "-c" "echo Instance i-0f04843e6faa8
cc7d with public IP 3.28.135.6 has been created\n"]
aws_instance.myapp-server (local-exec): Instance i-0f04843e6faa8cc7d with public IP 3.28.135.6 ha
s been created
aws_instance.myapp-server: Creation complete after 1m1s [id=i-0f04843e6faa8cc7d]

Apply complete! Resources: 1 added, 2 changed, 0 destroyed.

Outputs:

aws_instance_public_ip = "3.28.135.6"
@23-22411-027-max → ~/Lab12 $ terraform output
aws_instance_public_ip = "3.28.135.6"
```

⚠ Not secure 3.28.135.6

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task 6

```
GNU nano 7.2          entry-script.sh *
listen 443 ssl;
server_name $PUBLIC_IP;
ssl_certificate /etc/ssl/certs/selfsigned.crt;
ssl_certificate_key /etc/ssl/private/selfsigned.key;

location / {
    root /usr/share/nginx/html;
    index index.html;
    # proxy_pass http://158.252.94.241:80;
    # proxy_pass http://backend_servers;
}

server {
    listen 80;
    server_name _;
    return 301 https://$host$request_uri;
}
EOF

# Test and restart Nginx
systemctl restart nginx
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  bash - Lab12 + v [ ] [ ] ...

@23-22411-027-max → ~/Lab12 $ terraform apply -auto-approve
aws_instance.myapp-server: Still creating... [00m10s elapsed]
aws_instance.myapp-server: Creation complete after 13s [id=i-0c88ed25e94d07442]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

aws_instance_public_ip = "51.112.47.28"
@23-22411-027-max → ~/Lab12 $

aws_instance_public_ip = 51.112.47.28
● @23-22411-027-max → ~/Lab12 $ terraform output
aws_instance_public_ip = "51.112.47.28"
○ @23-22411-027-max → ~/Lab12 $
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task 7

```
GNU nano 7.2                                apache.sh *
```

```
#!/bin/bash
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd

echo "<h1>Welcome to My Web Server</h1>" > /var/www/html/index.html
hostnamectl set-hostname myapp-webserver
echo "<h2>Hostname: $(hostname)</h2>" >> /var/www/html/index.html

TOKEN=$(curl -s -X PUT "http://169.254.169.254/latest/api/token" \
-H "X-aws-ec2-metadata-token-ttl-seconds: 21600")

echo "<h2>Private IP: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" \
http://169.254.169.254/latest/meta-data/local-ipv4)</h2>" >> /var/www/html/index.html

echo "<h2>Public IP: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" \
http://169.254.169.254/latest/meta-data/public-ipv4)</h2>" >> /var/www/html/index.html

echo "<h2>Public DNS: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" \
http://169.254.169.254/latest/meta-data/public-hostname)</h2>" >> /var/www/html/index.html

echo "<h2>Deployed via Terraform</h2>" >> /var/www/html/index.html
```



```
data "http" "my_ip" {
  url = "https://icanhazip.com"
}

module "myapp-subnet" {
  source = "./modules/subnet"
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  env_prefix = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-web-1" {
  source = "./modules/webserver"
  env_prefix = var.env_prefix
  instance_type = var.instance_type
  availability_zone = var.availability_zone
  public_key = var.public_key
  my_ip = local.my_ip
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_id = module.myapp-subnet.subnet.id
  script_path = "./apache.sh"
  instance_suffix = "1"
  security_group_id = aws_default_security_group.default_sg.id
}
```

```
#output "aws_instance_public_ip" {
#  value = aws_instance.myapp-server.public_ip
#}

output "aws_instance_public_ip" {
  value = module.myapp-webserver.instance_public_ip
}

output "aws_web-1_public_ip" {
  value = module.myapp-web-1.aws_instance.public_ip
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash - Lab12 + v [] [] ... []

```
@23-22411-027-max →~/Lab12 $ terraform apply -auto-approve
aws_instance.myapp-server (remote-exec): Verifying : nginx-mimety 7/7

aws_instance.myapp-server (remote-exec): Installed:
aws_instance.myapp-server (remote-exec): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
aws_instance.myapp-server (remote-exec): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): libunwind-1.4.0-5.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): nginx-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch
aws_instance.myapp-server (remote-exec): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

aws_instance.myapp-server (remote-exec): Complete!
aws_instance.myapp-server (remote-exec): Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
aws_instance.myapp-server: Creation complete after 36s [id=i-064be5ecb06d4d7bd]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

aws_instance_public_ip = "158.252.85.83"
@23-22411-027-max →~/Lab12 $
```

```
The authenticity of host '51.112.52.48 (51.112.52.48)' can't be established.
ED25519 key fingerprint is SHA256:Qx/H0vBJfSDGIttGqg4REjqRq7JAUPOI/c08IbNU8d4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '51.112.52.48' (ED25519) to the list of known hosts.

#_
~\_####_ Amazon Linux 2023
~~~\_#####\
~~~\###|
~~~\#/____ https://aws.amazon.com/linux/amazon-linux-2023
~~~V~' '->
~~~~
~~~~_._/
~~~/_/_/_/
~~~/_m/'

[ec2-user@mvapp-webserver ~]$
```


PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ssh - Lab12

```
location / {  
#       root /usr/share/nginx/html;  
#       index index.html;  
#       proxy_pass http://<web-1-public-ip>:80;  
#       proxy_pass http://backend_servers;  
}
```

- nginx.service - The nginx HTTP and reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; **enabled**; preset: **disabled**)
Active: **active (running)** since Tue 2026-01-06 03:16:15 UTC; 12s ago
Process: 27197 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
Process: 27198 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
Process: 27199 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
Main PID: 27200 (nginx)
Tasks: 3 (limit: 1067)
Memory: 3.3M
CPU: 47ms
CGroup: /system.slice/nginx.service
├─27200 "nginx: master process /usr/sbin/nginx"
├─27201 "nginx: worker process"
└─27202 "nginx: worker process"

Jan 06 03:16:15 myapp-webserver systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server:

Jan 06 03:16:15 myapp-webserver nginx[27198]: nginx: [warn] could not build optimal type: 'uint16_t' (16 bits) for

Jan 06 03:16:15 myapp-webserver nginx[27199]: nginx: the configuration file /etc/nginx/nginx.conf was not found (2)

```
ec2-user@myapp-webserver ~]$ cat /var/log/nginx/error.log
026/01/06 03:04:47 [emerg] 26575#26575: bind() to 0.0.0.0:80 failed (98: Address already
in use)
026/01/06 03:04:47 [emerg] 26575#26575: bind() to [::]:80 failed (98: Address already in
e)
026/01/06 03:04:47 [notice] 26575#26575: try again to bind() after 500ms
026/01/06 03:04:47 [emerg] 26575#26575: bind() to 0.0.0.0:80 failed (98: Address already
in use)
026/01/06 03:04:47 [emerg] 26575#26575: bind() to [::]:80 failed (98: Address already in
e)
026/01/06 03:04:47 [notice] 26575#26575: try again to bind() after 500ms
026/01/06 03:04:47 [emerg] 26575#26575: bind() to 0.0.0.0:80 failed (98: Address already
in use)
026/01/06 03:04:47 [emerg] 26575#26575: bind() to [::]:80 failed (98: Address already in
e)
026/01/06 03:04:47 [notice] 26575#26575: try again to bind() after 500ms
026/01/06 03:04:47 [emerg] 26575#26575: bind() to 0.0.0.0:80 failed (98: Address already
in use)
026/01/06 03:04:47 [emerg] 26575#26575: bind() to [::]:80 failed (98: Address already in
e)
026/01/06 03:04:47 [notice] 26575#26575: try again to bind() after 500ms
026/01/06 03:04:47 [emerg] 26575#26575: still could not bind()
026/01/06 03:10:14 [warn] 26867#26867: could not build optimal types_hash, you should inc
ase either types_hash_max_size: 1024 or types_hash_bucket_size: 64; ignoring types_hash_t
ket_size
026/01/06 03:10:14 [emerg] 26867#26867: cannot load certificate "/etc/ssl/certs/selfsigne
```

```
[ec2-user@myapp-webserver ~]$ cat /var/log/nginx/access
[ec2-user@myapp-webserver ~]$
```

```
[ec2-user@myapp-webserver ~]$ cat /etc/nginx/mime.types
types {
application/A2L                a2l;
application/AML                aml;
application/andrew-inset      ez;
application/ATF               atf;
application/ATFX              atfx;
application/ATXML             atxml;
application/atom+xml          atom;
application/atomcat+xml       atomcat;
application/atomdeleted+xml   atomdeleted;
application/atomsvc+xml       atomsvc;
application/atsc-dwd+xml      dwd;
application/atsc-held+xml     held;
application/atsc-rsat+xml     rsat;
application/auth-policy+xml   apxml;
application/bacnet-xdd+zip     xdd;
application/calendar+xml      xcs;
application/cbor              cbor;
application/cccx              c3ex;
application/ccmp+xml          ccmp;
application/ccxml+xml         ccxml;
application/CDFX+XML          cdfx;
application/cdmia-capability  cdmia;
application/cdmia-container   cdmic;
application/cdmia-domain      cdmid;
application/cdmia-object      cdmio;
application/cdmia-queue       cdmioq;
application/CEA                cea;
application/cellml+xml         cellml cml;
application/clue_info+xml      clue;
application/cms                cmsc;
application/cpl+xml            cpl;
```

```
-----BEGIN CERTIFICATE-----
[ec2-user@myapp-webserver ~]$ cat /etc/ssl/certs/selfsigned.crt
-----BEGIN CERTIFICATE-----
MIIDCzCCAF0gAwIBAgIUe4hvraKMxtzcIidE/vIitW5pJG0wDQYJKoZIhvcNAQEL
BQAwFTETMBEGA1UEAwwKMTAuMC4xMC4xMjAeFw0yNjAxMDYwMzEyNTBaFw0yNzAx
MDYwMzEyNTBaMBUxEzARBgNVBAMMCjEwLjAuMTAuMTIwggEiMA0GCSqGSIb3DQEB
AQUAA4IBDwAwggEKAAoIBAQCzDFDPwA2Wu1xaKgDpmz0Aixt59th4Vqi20myurVJa
akCqiybmHjxMYUyQ5oakPk+DshkBya+jZRcMlz8J27cytSNUPqU6tVXNV+uMbEWt
E1R2qBFNxHKcazId3uqjQf6auzvBYUwhX3GCd/ADW5is7xjY7Qsos7w3m/64sMvn
Tls1ic9Q6pdaE753w01o454T8TWnfP+X4djp4WUwirFGXknajc7j9wRGhUB+W3jk
YY6zhcjfvzwR7lmm/DnoFxnrrjp4VBSQjudhBtVkkGOXoVNdbGwM5f4GYcPhpKbXx
fXzhq0wjyKkTBbyQQVA2tJXSCEcIEdK+CJJGoWBzYc6DAgMBAAGjUzBRMB0GA1Ud
DgQWBBS3EXw1bEbqVHFJ4L+IP9l0HEo8PDAfBgNVHSMEGDAWgBS3EXw1bEbqVHFJ
4L+IP9l0HEo8PDAPBgNVHRMBAf8EBTADAQH/MA0GCSqGSIb3DQEBEwUAA4IBAQCJ
8SSdCwfYkzqol5NoV6fbUkmCzAnskWro21ygNtmWEIQXyjf1dnEpXSJp4XTFn2DF
JHj/Lvc10Jhb8509+1z5E12SBXfTm3MVvXpUM4cm7vWMYEdT+ha5dzcK1d6TKbkH
5oynmcXYR9fPwe+VFNgjyhrztShCStrbk/E/bEZBg95GTp4oNR5s/jZJTD24SFTg
Avvlp731feUBD6+r4A4TXqK81W5EWSTRhgImism2zg6F5jhk3iVaYVUfXN8s8qgT
H+5UE0w4V1W0XbHIEtcFmmeuxmOIQYmgsgnGmDIJcnJ3tT0L9zg6lpr7wB0dNtH6
2xWH9ogoDyz4keqGhWGN
-----END CERTIFICATE-----
[ec2-user@myapp-webserver ~]$
```



```
-----BEGIN PRIVATE KEY-----
MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBAcwggSjAgEAAoIBAQCzDFDPwA2Wulxa
KgDpmz0Aixt59th4Vqi2OmyurVJaakCqiybmHjxMYUyQ5oaKPk+DshkBya+jZRcM
lz8J27cytSNUPqU6tVXNV+uMbEWtE1R2qBFNxHKcazId3uqjQf6auzvBYUwhX3GC
d/ADW5is7xjY7Qsos7w3m/64sMvnTls1ic9Q6pdaE753w01o454T8TwnfP+X4djP
4WUwirFGXknaic7j9wRGhUB+W3jkYY6zhcjfvzwR7lmm/DnoFxnrrjp4VBSQjudhB
tVkkGOXoVNdbGwM5f4GYcPhPKbXxfXzhqOwjyKkTBbyQQVA2tJXSCEcIEdK+CJJG
oWBZyC6DAgMBAAECggEAAeNIWthojEVMwfqdZdkoLIBK+t86axJm1v0ItlLHgoKK
SgILl/OLqTaBaymzE3QsVi29K8Knom5nbs+Nb5osdVFpnPLIJTVSQUcDCcj+dx00
GUguXoaKutF6hswZ5zzD5XFb6okPF279MGX5Vt+5XWFr/qWJL5Wtj1yKtAHRNJAF
asPrnsB1Dr5gnCljJ8JDtLV6e4cCESb2gAu9HFUP5DyWuyFPZwV3Q07er6CtiQkh
sharkMinEF9MvDm3YcFwe6hASPM8FgU9iCyH13ZK4S4Y3HtmG7KXPxt7+2zFupya
B/GP7R6VYe0wHKpbj0aaLdBaitnFpfvbu4HAmHzAAQKBgQDhZwgLCiDiGtUF1J7F
aSxb/cD9mzaz9zr8jDtEp2QFQfGgu/TxtyNwPMP1HnV5KC5wkJSY/5CSwEc6MXKy
rJ8sSvQIQf2QRBjVKK4gxn4dAyi3HvUiGvWCpgsMDgUKjItKlwh8D/GXPCpQaHb7
dGyDSoqnC31tcnJbL+Ifxe2gAQKBgQDLWmwi3q25VmyHiNb753qyuMJ8lYo+0yE/
LbPGQHiafoEMgvs2tTYF/1YPPgzWt+CZy1+3rDwOFGOn+ITbhr3cLdqynhsn0qvA
0GQ0Swfy9F7dtHTxYnKtLuK9taF2p0qMfGvg0hOkX4mL4rUhZfj9/7LXe4l3igJ/
8AmuQAjugwKBgBYhyr1iwjNDxzTqWqihmhRyRxffvLBRDz6q+krUcNK/Vd1ldzig
td8YaFe/90Lvmhe9aonXVjfCn3vMTSnUH1D3Dx0mgwFeXbp04xwewVVR8RaB+Knf
ocmxShZl7gLKKAcr605bNmQWIsWAbGZmWR1PHWwniapMeYQqoJBkG+ABAoGBAMhJ
njpWAtuKAMkZgU2I970yAHS+wPN8P8m/VNLOSzs3oBlXpOCEaNPXFEU94Zag2pX
Akag0GG5uVIGl6y0sKl9TGsjCHGB+wqk9hz/TawA/rL0+z2IjpR+mnrpu0d1NVL7
dTFxEGEMi6gIughULILX5ykPlxzzBHVIUfCbR3DLAoGAPddUbqPTF/8TP3orUh1K
MupMjN8W06kbsDXqbfckQ6iY2Vut+J0rdv6p770ALmCuSAbLceTYSHMrXtn11bA
AdVrm+6XtJHsY0gxb+DMIZkaBSLfZsY3bpgHbp6niw4UDeKV8AQJ3OMlSp/Q3y1d
MkV5FCi70B2Daklwpr8CXTc=
-----END PRIVATE KEY-----
[ec2-user@myapp-webserver ~]$
```

Welcome to My Web Server

Hostname: myapp-webserver

Task 8

```
}

module "myapp-web-1" {
  source          = "./modules/webserver"
  env_prefix      = var.env_prefix
  instance_type   = var.instance_type
  availability_zone = var.availability_zone
  public_key      = var.public_key
  my_ip           = local.my_ip
  vpc_id          = aws_vpc.myapp_vpc.id
  subnet_id       = module.myapp-subnet.subnet.id
  script_path     = "./apache.sh"
  instance_suffix = "1"
  security_group_id = aws_default_security_group.default_sg.id
}

module "myapp-web-2" {
  source = "./modules/webserver"
  env_prefix = var.env_prefix
  instance_type = var.instance_type
  availability_zone = var.availability_zone
  public_key = var.public_key
  my_ip = local.my_ip
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_id = module.myapp-subnet.subnet.id
  script_path = "./apache.sh"
  instance_suffix = "2"
}
```

```

GNU nano 7.2                                outputs.tf *
#output "aws_instance_public_ip" {
#  value = aws_instance.myapp-server.public_ip
#}

output "aws_instance_public_ip" {
  value = module.myapp-webserver.instance_public_ip
}

#output "aws_web-1_public_ip" {
#  value = module.myapp-web-1.aws_instance.public_ip
#}

output "aws_web_1_public_ip" {
  value = module.myapp-web-1.instance_public_ip
}

output "aws_web-2_public_ip" {
  value = module.myapp-web-2.aws_instance.public_ip
}

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

upstream myapp {
    server 158.252.77.178:80;
    server 158.252.79.111:80;
}

server {
    listen 80;
    location / {
        proxy_pass http://myapp;
    }
}
~
~
~
~

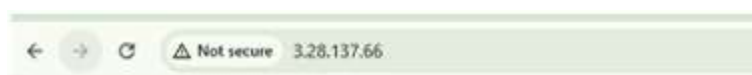
```



Hello from Terraform



Welcome to Server ONE



Welcome to Server TWO

Task 9

```
GNU nano 8.3 /etc/nginx/nginx.conf
# For more information on configuration, see:
#   * Official English Documentation: http://nginx.org/en/docs/
#   * Official Russian Documentation: http://nginx.org/ru/docs/

user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

# Load dynamic modules. See /usr/share/doc/nginx/README.dynamic.
include /usr/share/nginx/modules/*.conf;

events {
    worker_connections 1024;
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';
```


← → ↻ ⚠ Not secure 3.28.137.66

Welcome to Server ONE

GNU nano 8.3

/etc/nginx/nginx.conf

```
access_log /var/log/nginx/access.log main;

upstream myapp {
    server 158.252.77.178:80 backup;
    server 158.252.79.111:80;
}

server {
    listen 80;
    location / {
        proxy_pass http://myapp;
    }
}

sendfile on;
tcp_nopush on;
keepalive_timeout 65;
types_hash_max_size 4096;

include /etc/nginx/mime.types;
default_type application/octet-stream;
```

← → ↻ ⚠ Not secure 3.28.137.66

Welcome to Server TWO

Task 10

```
http {
    proxy_cache_path /var/cache/nginx levels=1:2 keys_zone=my_cache:10m inactive=60m max_size=1g;

    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;
```

▼ Response Headers ☐ Raw	
Accept-Ranges	bytes
Connection	keep-alive
Content-Length	1532
Content-Type	text/html; charset=UTF-8
Date	Wed, 31 Dec 2025 14:35:07 GMT
Etag	"5fc-6472fb4e1feb7"
Last-Modified	Tue, 30 Dec 2025 18:41:24 GMT
Server	nginx/1.28.0
Upgrade	h2,h2c
X-Cache-Status	MISS

Name	✕ Headers	Preview	Response	Initiator	Timing
3.28.137.66	Referrer Policy		strict-origin-when-cross-origin		
	▼ Response Headers ☐ Raw				
	Connection		keep-alive		
	Date		Mon, 05 Jan 2026 18:43:34 GMT		
	Etag		"1f-647a759633ca9"		
	Last-Modified		Mon, 05 Jan 2026 17:25:45 GMT		
	Server		nginx/1.28.0		
	X-Cache-Status		HIT		
	▼ Request Headers ☐ Raw				
1 requests	205 B transferred	3 Accept	text/html,application/xhtml+xml,application/		

Cleanup:

```
@23-22411-027-max →~/Lab12 $ terraform destroy
```

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
aws_route_table_association.myapp_rta: Destroying... [id=rtbassoc-0f2094bcea12cbcf6]
aws_instance.myapp-server: Destroying... [id=i-0f04843e6faa8cc7d]
aws_route_table_association.myapp_rta: Destruction complete after 1s
aws_route_table.myapp_rt: Destroying... [id=rtb-09b1a4a1d88fdaa4b]
aws_route_table.myapp_rt: Destruction complete after 0s
aws_internet_gateway.myapp_igw: Destroying... [id=igw-0a8dbb2ec2adaa766]
aws_instance.myapp-server: Still destroying... [id=i-0f04843e6faa8cc7d, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0a8dbb2ec2adaa766, 00m10s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-0f04843e6faa8cc7d, 00m20s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0a8dbb2ec2adaa766, 00m20s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-0f04843e6faa8cc7d, 00m30s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0a8dbb2ec2adaa766, 00m30s elapsed]
aws_internet_gateway.myapp_igw: Destruction complete after 38s
aws_instance.myapp-server: Still destroying... [id=i-0f04843e6faa8cc7d, 00m40s elapsed]
aws_instance.myapp-server: Destruction complete after 41s
aws_subnet.myapp_subnet_1: Destroying... [id=subnet-0a1fb953af1d0679f]
aws_key_pair.ssh-key: Destroying... [id=serverkey]
aws_security_group.myapp_sg: Destroying... [id=sg-0248bdd0bf31447a5]
aws_key_pair.ssh-key: Destruction complete after 0s
aws_subnet.myapp_subnet_1: Destruction complete after 0s
aws_security_group.myapp_sg: Destruction complete after 0s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0cda72e87363cd088]
aws_vpc.myapp_vpc: Destruction complete after 1s
```

Destroy complete! Resources: 8 destroyed.

LAB-12 [CODESPACES: FRIENDL...]

- Lab12
 - .terraform
 - modules
 - modules.json
 - providers/registry.terraform.io/has...
 - aws/5.100.0/linux_amd64
 - LICENSE.txt
 - terraform-provider-aws_v5.100.0_x5
 - http
 - modules
 - subnet
 - main.tf
 - outputs.tf
 - variables.tf
 - webserver
 - main.tf
 - outputs.tf
 - variables.tf
 - .terraform.lock.hcl
 - main.tf
 - outputs.tf
 - terraform.tvarfs

PROBLEMS OUTPUT DEBUG CONSOLE

@23-22411-027-max -> /worksp